

On the Number of Branched Covers of an Elliptic Curve

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1 Introduction

Let E be an elliptic curve over $\mathbb{C}$, and $g \geq 1, m \geq 2$ integers.

A pair (C, f) consisting of a smooth complex curve C and a holomorphic map $f : C \rightarrow E$ is called a *simple m -branched cover* if C is connected, every ramification point has index m , and different ramification points have different images.

Let $X_{g,d}^{(m)}$ be the set of isomorphism classes of m -branched covers of E of degree d and genus g , and let

$$N_{g,d}^{(m)} = \sum_{(C,f) \in X_{g,d}^{(m)}} \frac{1}{|Aut(C, f)|}.$$

Define

$$F_g^{(m)}(q) = \sum_{d \geq 1} N_{g,d}^{(m)} q^d,$$

for $g \geq 2$. The case $g = 1$ must be treated separately, since we have to consider the contribution of constant maps. We define

$$F_1(q) = -\frac{1}{24} \log q + \sum_{d \geq 1} N_{1,d}^{(m)} q^d.$$

In this document will present an sketch of the proof the case $m = 2$ of the next theorem; for the general case we refer to [3].

Theorem 1 (*Dijkgraaf $m = 2$ [1], Ochiai $m \geq 2$ [3]*) For $g \geq 2$, $F_g^{(m)}(q)$ is a quasimodular form for the full modular subgroup $SL(2, \mathbb{Z})$. In the case $m = 2$, $F_g(q)$ has weight $6g - 6$.

From now on we will try to omit $m = 2$.

2 Quasimodular forms

Let's recall what a quasimodular form is. Let $Y(\tau) = 4\pi Im(\tau)$. Kaneko and Zagier define an *almost homomomorphic modular form of weight k* to be a function on the upper half plane of the form

$$\sum_{m=0}^n f_m(\tau) Y^{-m},$$

where each f_m is holomorphic and grows polynomially in $1/Y$ as $Y \rightarrow \infty$, and such that F satisfies the usual weight k identity for the full modular group $SL(2, \mathbb{Z})$. A quasimodular form of degree k is defined to be the coefficient $f_0(\tau)$ in a holomorphic modular form of weight k .

It can be proved that the graded algebra of quasimodular forms is $\mathbb{C}[E_2, E_4, E_6]$, with $deg(E_i) = i$. Two different approaches to this can be found in [2], [4] and [5]. So theorem 1 asserts that $F_g^{(m)}(q)$, for $g, m \geq 2$ can be written as polynomials in E_2, E_4 and E_6 . Some explicit computations can be found in [6].

3 Partition functions

One of the steps in the proof of theorem 1 is to allow possibly disconnected covers of E , so let $\hat{X}_{g,d}$ and $\hat{N}_{g,d}$ be defined as in the introduction, but allowing possibly disconnected smooth curves C . $\hat{N}_{g,d}$ and $N_{g,d}$ are finite by proposition 2 in the next section. Now we define partition functions for the connected and possibly disconnected case, respectively, by

$$Z(q, \lambda) = \sum_{g \geq 1} \sum_{d \geq 1} \frac{N_{g,d}}{(2g-2)!} q^d \lambda^{2g-2} = \sum_{g \geq 1} \frac{F_g(q)}{(2g-2)!} \lambda^{2g-2},$$

and

$$\hat{Z}(q, \lambda) = \sum_{g \geq 1} \sum_{d \geq 1} \frac{\hat{N}_{g,d}}{(2g-2)!} q^d \lambda^{2g-2}.$$

The strategy is to relate this two partitions functions by means of the next proposition.

Proposition 1 $\hat{Z}(q, \lambda) = \exp(Z(q, \lambda)) - 1$.

The proof of it is stricky combinatorial and can be considered as a consequence of the exponential formula in combinatorics. See, for example, corollary 3.4.1. in [7].

4 Reducing $\hat{N}_{g,d}$ to the trace on an operator

Let us choose b different points $P_1, \dots, P_b$ in E , where $b = 2g - 2$, that is, by Hurwitz formula, the number of ramification points of a 2-branched cover of E , and let π_1^b be the fundamental group of the b -punctured elliptic curve $E - P_1, \dots, P_b$. Then

$$\pi_1^b = \langle \gamma_1, \dots, \gamma_b, \alpha, \beta \mid \gamma_1 \cdots \gamma_b = \alpha \beta \alpha^{-1} \beta^{-1} \rangle$$

Let c_{trans} be the conjugation class of the transpositions in S^d . Define

$$\Phi_{g,d} = \{ \phi \in \text{Hom}(\pi_1^b, S_d) \mid \phi(\gamma_i) \in c_{trans} \text{ for every } i \}.$$

Then S_d acts on $\Phi_{g,d}$ by conjugation.

By the next proposition, computing $\hat{N}_{g,d}$ is equivalent to computing $|\Phi_{g,d}|$.

Proposition 2 *a. Monodromy induces a bijection $\hat{X}_{g,d} \cong \Phi_{g,d}/S_d$;*

b. $\hat{N}_{g,d} = |\Phi_{g,d}|/d!$.

We want now want to compute $|\Phi_{g,d}|$. Since we have a description of π_1^b by generators and relations, we get a bijective correspondence between $\Phi_{g,d}$ and the set

$$\{ (\tau_1, \dots, \tau_b, \tau, \sigma) \in (S_d)^{b+2} \mid \tau_1 \cdots \tau_b = \tau \sigma \tau^{-1} \sigma^{-1}, \tau_i \in c_{trans} \text{ for every } i \}.$$

So we need to count the elements of this set. Fix $\sigma \in S_d$. Let

$$P_{g,d}(\sigma) = \{ (\tau_1, \dots, \tau_b) \in (S_d)^b \mid \tau_1 \cdots \tau_b \sigma \text{ and } \sigma \text{ are conjugate, } \tau_i \in c_{trans} \text{ for every } i \}.$$

Since given a τ such that $\tau_1 \cdots \tau_b = \tau \sigma \tau^{-1} \sigma^{-1}$ we can get all the other τ with such a property by multiplying it with an element of the conjugation class of σ , $c(\sigma)$, we have

$$|\Phi_{g,d}| = \sum_{\sigma \in S_d} \frac{d!}{|c(\sigma)|} |P_{g,d}(\sigma)|.$$

Since $P_{g,d}(\tau \sigma \tau^{-1}) = \tau P_{g,d}(\sigma) \tau^{-1}$, $|P_{g,d}| : S_d \rightarrow \mathbb{N}$ is a class function, therefore

$$|\Phi_{g,d}| = \sum_{\sigma \in \mathcal{C}_d} d! |P_{g,d}(\sigma)|,$$

and $\hat{N}_{g,d} = \sum_{c \in \mathcal{C}_d} |P_{g,d}(c)|$.

For now on, let's fix a complete set of representatives $\{\sigma_1, \dots, \sigma_m\}$ of the conjugation classes $\mathcal{C}_d = \{c_1, \dots, c_m\}$, $c(\sigma_i) = c_i$. Observe that m is the number of partitions of d (that we call $part(d)$) since the conjugacy class of an element of S_d is determined by the partition induced by its cycle decomposition. Consider the matrix

$$(M_d)_{ij} = |\{\tau \in c_{trans} \mid \tau\sigma_i \in c_j\}|.$$

An straightforward induction argument, with the base case being the last definition gives us

$$(M_d^k)_{ij} = |\{(\tau_1, \dots, \tau_k) \in (S_d)^k \mid \tau_1, \dots, \tau_k\sigma_i \in c_j, \tau_l \in c_{trans} \text{ for every } l\}|.$$

From this observation we get $(M_d^b)_{i,i} = |P_{g,d}(\sigma_i)|$ and therefore we have $\hat{N}_{g,d} = Tr(M_d^b)$. We also conclude that $(M_d^k)_{i,i} = 0$ for odd k . Therefore we have the following expression for $\hat{Z}$:

$$\hat{Z}(q, \lambda) = \sum_{g \geq 1} \sum_{d \geq 1} \frac{Tr(M_d^b)}{(2g-2)!} q^d \lambda^{2g-2} = \sum_{d \geq 1} (\exp(M_d \cdot \lambda)) q^d.$$

Let $\alpha_{d,i}$ be the eigenvalues of M_d . Then $\hat{N}_{g,d} = \sum_{i=1}^m \alpha_{d,i}^k$. So the last expression gives us the following proposition.

Proposition 3 $\hat{Z}(q, \lambda) = \sum_{d \geq 1} \sum_{i \geq 1}^{part(d)} \exp(\alpha_{d,i} \lambda) q^d$.

5 Diagonalizing M_d

Recall that the class algebra $\mathcal{H}_d$, which is the center of the group algebra $\mathbb{C}[S_d]$ (i.e., functions $S_d \rightarrow \mathbb{C}$ constant on conjugation classes) has a basis $(z_c)_{c \in \mathcal{C}_d}$, where $z_c = \sum_{\sigma \in c} \sigma$. Since the inverse of a transposition is a transposition, we have the following easy proposition.

Proposition 4 *The matrix of the operator on $\mathcal{H}_d$ defined by multiplication by $z_{c_{trans}}$ is the transpose of M_d .*

Now, the eigenvalues of this operator follow from Schur orthogonality relations:
For any irreducible character χ of S_d , define

$$w_\chi = \frac{\dim(\chi)}{d!} \sum_{c \in \mathcal{C}_d} \chi(c^{-1}) z_c = \frac{\dim(\chi)}{d!} \sum_{\sigma \in \mathcal{C}_d} \chi(\sigma^{-1}) \sigma.$$

Schur orthogonality relations tell us that $w_\chi \cdot w_{\chi'} = \delta_{\chi, \chi'} w_\chi$ (this is Kronecker's delta), and

$$z_c = \sum_{\chi} \left(\frac{|c^{-1}| \chi(c^{-1})}{\dim(\chi)} \right) w_\chi.$$

From this identities it follows immediately that $(w_\chi)_{\chi \in \mathcal{R}_d}$ is basis of $\mathcal{H}_d$ which diagonalizes multiplication by $z_{c_{trans}}$. The eigenvector w_χ has eigenvalue

$$\frac{\binom{d}{2} \chi(c)}{\dim(\chi)}.$$

6 Frobenius formula and Haneko-Zagier result

It is known that irreducible representations of S_d are in one-to-one correspondence with partitions of d , (for example, via the Sprech representation). The particular construction of this representations will not be relevant to us, but a fomula of Frobenius will. Let χ be an irreducible representation of S_d and $\lambda = (\lambda_1, \dots, \lambda_l)$ its corresponding partition. Define $u_i = \lambda_i - i + \frac{1}{2}$ and v_i defined as for u_i , but by considering the transpose partition of λ . The Frobenius formula we will use affirms that

$$\frac{\binom{d}{2} \chi(c)}{\dim(\chi)} = \frac{1}{2} \sum_i (u_i^2 - v_i^2),$$

so it computes the eigenvalues found in the last section.

Consider the product

$$\prod_{u \in \mathbb{N} + \frac{1}{2}} (1 + \zeta q^u e^{\frac{u^2}{2}\lambda}) \prod_{v \in \mathbb{N} + \frac{1}{2}} (1 + \zeta^{-1} q^v e^{\frac{v^2}{2}\lambda})$$

We can see it as a Laurent formal series in ζ . Propositions 3 and 4 gives us

Proposition 5 $\hat{Z}(q, \lambda) = \text{coef. of } \zeta^0 \text{ in } \prod_{u \in \mathbb{N} + \frac{1}{2}} (1 + \zeta q^u e^{\frac{u^2}{2}\lambda}) \prod_{v \in \mathbb{N} + \frac{1}{2}} (1 + \zeta^{-1} q^v e^{\frac{v^2}{2}\lambda}) - 1.$

In [2], Kaneko and Zagier considered the series

$$\Theta(q, \lambda, z) = \prod_{n \geq 1} (1 - q^n) \prod_{u \in \mathbb{N} + \frac{1}{2}} (1 + \zeta q^u e^{\frac{u^2}{2}\lambda}) \prod_{v \in \mathbb{N} + \frac{1}{2}} (1 + \zeta^{-1} q^v e^{\frac{v^2}{2}\lambda}).$$

Let $\Theta_0(q, \lambda)$ be the coefficient of ζ^0 in $\Theta(q, \lambda, z)$. Then, since $\Theta_0(q, \lambda) = \Theta_0(q, -\lambda)$, we have an expansion

$$\Theta_0(q, \lambda) = \sum_{n \geq 0} A_n(q)^{2n}.$$

Kaneko and Zagier prove (theorem 1, [2]) that $A_n(q)$ is a quasimodular form of weight $6n$. This implies that the coefficient of λ^{2g-2} in $\log \Theta_0(q, \lambda)$ is a modular form of weight $6g - 6$, since, by using the taylor series of the logarithm around -1 , we have

$$\log \Theta_0(q, \lambda) = \log(1 + \sum_{n \geq 1} A_n(q)^{2n}) = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} (\sum_{n \geq 1} A_n(q)^{2n})^m.$$

On the other hand, by propositions 1 and 5, we get

$$\log \Theta_0(q, \lambda) = \log(\prod_{n \geq 1} (1 - q^n)(\hat{Z}(q) + 1)) = \sum_{n \geq 1} \log(1 - q^n) + \log(\hat{Z}(q, \lambda) + 1) = \sum_{n \geq 1} \log(1 - q^n) + Z(q, \lambda).$$

Since the coefficient of λ^{2g-2} in this last sum is the one coming from $Z(q, \lambda)$, i.e.,

$$\frac{F_g(q)}{(2g-2)!},$$

the proof of theorem 1 for $m = 2$ is concluded.

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